

Università degli Studi di Padova

DIPARTIMENTO DI INGEGNERIA DELL'INFORMAZIONE

UNIVERSITY OF PADOVA

Department of Information Engineering

Bachelor's Degree in Computer Engineering

Final examination report

Computing the median with limited memory: study and experimentation of the multi-pass algorithm by Munro and Paterson

SUPERVISOR

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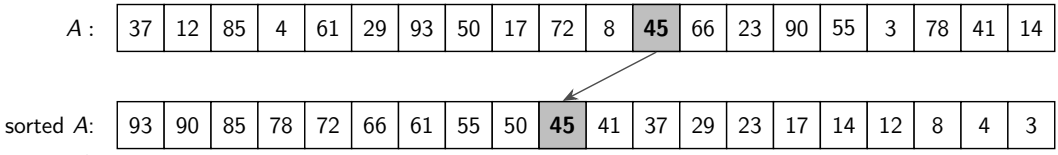
CANDIDATE

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The selection problem

- **Input**: sequence $A = (a_1, \ldots, a_N)$ of N distinct elements from a totally ordered domain
- **Rank**: $a_{(k)}$ is the k-th **largest** element of A
- **Selection**: given $1 \leq k \leq N$, find $a_{(k)}$ without sorting the sequence
- **Median**: the case $k = \lceil N/2 \rceil$



$N = 20$, median: $k = \lceil 20/2 \rceil = 10$, hence $a_{(10)} = 45$

The SELECT algorithm

- **SELECT**: deterministic, *median of medians*
  - groups of 5 elements and the median of each group
  - pivot x: the median of the medians
  - Partition around x and recursion on the part containing the sought element
- $O(N)$ time even in the worst case, but with the whole input in memory

How can the median be found when the available memory is limited instead?

Goal of the thesis

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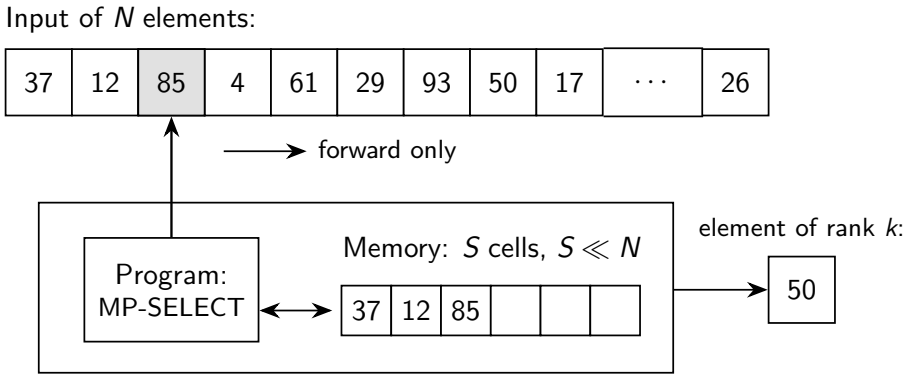
Study of the multi-pass algorithm **MP-SELECT** by Munro and Paterson for selection with limited memory, and its **experimental analysis**.

- S: available memory cells, with $S \ll N$; each cell holds one element
- P: number of **passes**, i.e. complete readings of the input; it is the cost measure
- **Candidates**: elements lying between two filters ($f_{\min}$ and $f_{\max}$) that may still be $a_{(k)}$
- n_p: number of candidates **in pass** $p \in [1, P]$, which decreases from pass to pass

Qualitative trade-off: less memory $S \implies$ more passes P

J. I. Munro, M. S. Paterson, *Selection and sorting with limited storage*, Theoretical Computer Science 12 (1980) 315–323.

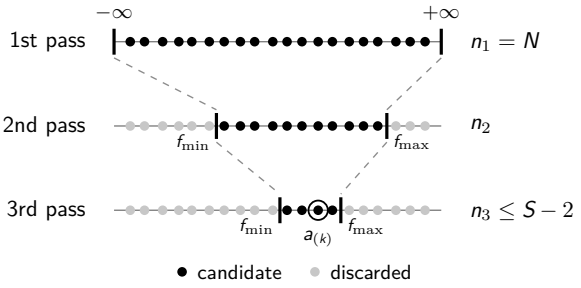
The model of computation: streaming



Streaming model (multi-pass): the input is read only sequentially, from start to end, possibly several times; at most S elements are kept in memory

RAM model (classical): the whole input is in memory, with direct access to any element

The multi-pass algorithm



In each pass:

1. **Reading**: the input is scanned once
2. **Filters**: only the candidates, between $f_{\min}$ and $f_{\max}$, are kept
3. **Sampling**: the candidates, read in sorted blocks, are halved several times (one out of two) and merged into a sample that fits in memory
4. **New filters**: from the sample, two closer filters that still enclose $a_{(k)}$

If the bound on the candidates is $\leq S - 2 \implies$ **one last pass** loads them all into memory and finds the sought element.

Goals of the experimental analysis

Bound on the candidates $\bar{n}_p$: the algorithm does not know n_p exactly, since counting it would require another pass. It therefore uses an upper bound $\bar{n}_p \geq n_p$, computed without reading the data, to:

- guarantee that the sample always fits in the S cells
- decide when to stop: if $\bar{n}_p \leq S - 2 \Rightarrow$ **last pass**

Experiments on random permutations of $1, \ldots, N$, for the median $k = \lceil N/2 \rceil$:

1. compare the number of passes P as S and N vary
2. measure how much $\bar{n}_p$ overestimates n_p
3. observe how the number of candidates evolves as the passes increase

Implementation

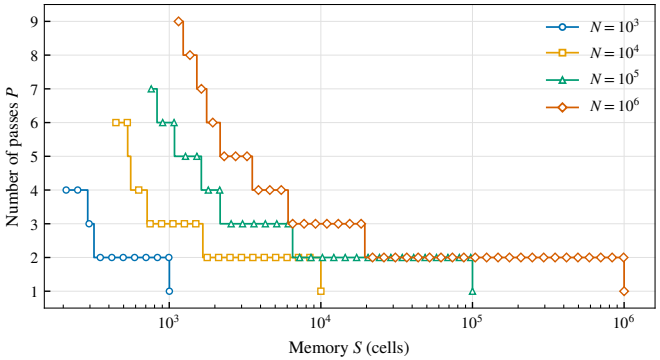
MP-SELECT algorithm implemented in **Python 3**

Results and plots produced with *Matplotlib*



https://github.com/ill-barba/bsc-thesis-unipd-median-selection

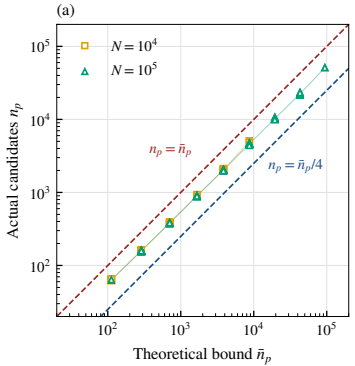
Trade-off between memory and passes

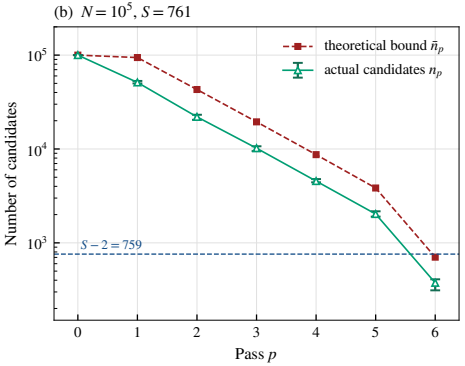


- Memory for P passes: $S \sim N^{1/P}$
- Logarithmic dependence on N, with $S = \Omega(\log^2 N)$
- Can be read both ways: $S \to P$ and $P \to S$

Theoretical bound: $\bar{n}_{p+1} = O\left(\frac{\bar{n}_p (\log \bar{n}_p)^2}{S}\right)$ **For fixed S:** $P \approx \frac{\log N}{\log S}$

Actual candidates and theoretical bound



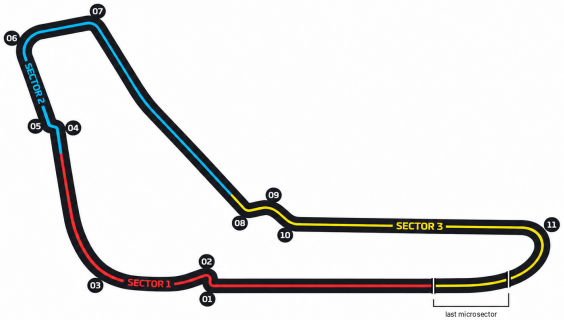


- $k = \lceil N/10 \rceil$ similar to the median; extreme ranks such as $k = 1$ have fewer candidates
- Bars initially close together due to averaging out
- "Slowdown" due to $\bar{n}_p$ driving the algorithm

Limitation: conservative bound (about twice the actual candidates), which can cost at most one more pass than necessary.

Possible development: an *ad hoc* greedy heuristic to attempt an early termination.

Identifying the thermal window



Last microsector $\sim 3.8\,\mathrm{s}$

Preparation lap j:

$$T_j = (t_{j,1}, \ldots, t_{j,N_j}), \qquad k_j = \left\lceil \frac{N_j}{2} \right\rceil$$

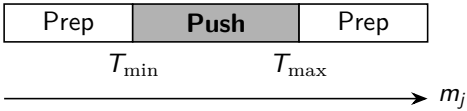
Median of the lap:

$$m_j = \text{MP-SELECT}(T_j, k_j, S)$$

First lap in the thermal window:

$$j^* = \min\{\, j \;:\; T_{\min} \le m_j \le T_{\max} \,\}$$

Graphically:



Extension to four tyres

One median for each tyre: $m_j^{FL},\ m_j^{FR},\ m_j^{RL},\ m_j^{RR}$

1. Conservative: Push only if all four are in the window

$$T_{\min} \leq m_j^w \leq T_{\max} \qquad \forall\, w \in \{FL, FR, RL, RR\}$$

2. Mean of the medians: more permissive

$$\overline{m}_j = \tfrac{1}{4}(m_j^{FL} + m_j^{FR} + m_j^{RL} + m_j^{RR}), \quad T_{\min} \leq \overline{m}_j \leq T_{\max}$$

3. Weighted mean: trade-off, different weights according to the stress on each tyre

$$\widetilde{m}_j = \alpha_{FL} m_j^{FL} + \alpha_{FR} m_j^{FR} + \alpha_{RL} m_j^{RL} + \alpha_{RR} m_j^{RR}$$
$$T_{\min} \leq \widetilde{m}_j \leq T_{\max}$$



Thermal window for each compound